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# 5. Data Warehousing and Online Analytical Processing

Knowledge Discovery in Databases

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# Outline



1. Data Warehouse: Basic Concepts

2. Data Warehouse Modeling: Data Cube and OLAP

3. Data Warehouse Design and Usage

4. Data Warehouse and Data Mining

5. Summary

6. Appendix

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# Data Warehouse: Basic Concepts

# What is a Data Warehouse?



William “Bill” H. Inmon is commonly referred to as the “father of the data warehouse”.

## Data Warehouse

A data warehouse is a **subject-oriented**, **integrated**, **time-variant**, and **nonvolatile** collection of data in support of management’s decision-making process.[1] Common abbreviations: DW or DWH.

**Other definitions exist:**

- “A data warehouse is a system that extracts, cleans, conforms, and delivers source data into a dimensional data store and then supports and implements querying and analysis for the purpose of decision making.”[2]
- A **decision-support** database that is **maintained separately** from the organization’s operational database.
- Supports information processing by providing a solid platform of **consolidated, historical data** for analysis.

**Data warehousing:** The process of constructing and using data warehouses.

[1] W. H. Inmon, *Building the Data Warehouse*. Wiley, 2005, 4th edition, ISBN: 978-076459-944-6
[2] R. Kimball and J. Caserta, *The Data Warehouse ETL Toolkit: Practical Techniques for Extracting, Cleaning, Conforming, and Delivering Data*. Wiley, 2004, ISBN: 978-0764567575

# Data Warehouse – Subject-oriented



- **Organized around major subjects.** Such as customer, product, sales.
- **Focusing on the modeling and analysis of data for decision makers.** Not on daily operations or transaction processing.
- **Provide a simple and concise view around particular subject issues.** By excluding data that are not useful in the decision-support process.

# Data Warehouse – Integrated



- **Constructed by integrating multiple heterogeneous data sources.**
  - Relational databases, flat files, online transaction records, . . .
- **Data-cleaning and data-integration techniques are applied.**
  - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources.
  - E.g., hotel price: currency, tax, breakfast covered.
  - When data is moved to the data warehouse, it is converted.
  - ETL – Extract, Transform, Load.

# Data Warehouse – Time-Variant



- **The time horizon for a data warehouse is significantly longer than that of operational systems.**
  - Operational database: current-value data.
  - Data warehouse: provide information from a historical perspective, e.g. past 5 — 10 years.
- **Every key structure in the data warehouse contains an element of time, explicitly or implicitly.**
- The key of operational data may or may not contain a "time element."

# Data Warehouse – Nonvolatile



- **A physically separate store of data.**
  - Transformed from the operational environment.
  - By copying.
- **No operational update of data:**
  - Hence, does not require transaction processing, i.e. no logging, recovery, concurrency control, etc.
  - Requires only three operations:
    1. Initial loading of data.
    2. Refresh (update, often periodically, e.g. over night).
    3. Access of data.

# Data Warehouse Usage



**Three kinds of data warehouse applications.**

1. **Information processing.**
   - Supports querying, basic statistical analysis, and reporting using crosstabs, tables, charts and graphs.
2. **Analytical processing.**
   - Multidimensional analysis of data warehouse data.
   - Supports basic OLAP operations such as slicing, dicing, drilling, and pivoting.
3. **Data mining.**
   - Knowledge discovery from hidden patterns.
   - Supports associations, constructing analytical models, performing classification and prediction, and presenting the mining results using visualization tools.

# OLTP vs. OLAP

| | OLTP | OLAP |
|---|---|---|
| **Users** | clerk, IT professional | knowledge worker |
| **Function** | day-to-day operations | decision support |
| **DB Design** | application-oriented | decision support |
| **Data** | current, up-to-date; detailed, flat relational; isolated | historical; summarized, multidimensional, integrated, consolidated |
| **Usage** | repetitive | ad-hoc |
| **Access** | read/write; index/hash on primary key | lots of scans |
| **Unit of Work** | short, simple transaction | complex query |
| **#-Records Accessed** | 10 | $10^6$ |
| **#-Users** | 1000 | 100 |
| **DB Size** | 100 MB to GB | 100 GB to TB |
| **Quantification** | transaction throughput | query throughput, response |

OLTP = Online Transaction Processing, OLAP = Online Analytical Processing

# Why a Separate Data Warehouse?



**High performance for both systems:**

- **DBMS**: tuned for OLTP; Access methods, indexing concurrency control, recovery.
- **Data Warehouse**: tuned for OLAP; Complex OLAP queries, multidimensional view, consolidation.

**Different functions and different data:**

- *Missing data (DBMS):* Decision support (DS) requires **historical data** which operational DBs do not typically maintain.
- *Data consolidation (warehouse):* DS requires **consolidation** (aggregation, summarization) of data from heterogeneous sources.
- *Data quality (warehouse):* Different sources typically use inconsistent data representations, codes and formats which have to be reconciled.

**Note**

There are more and more systems which perform OLAP analysis directly on relational databases.

# Three Data Warehouse Models



1. **Enterprise Warehouse:**
   - Collects all of the information about subjects spanning the entire organization.
2. **Data Mart:**
   - A **subset** of corporate-wide data that is of value to a **specific group of users**.
   - Typically contains (highly) summarized data.
   - Independent vs. dependent (directly from warehouse) data mart.
3. **Virtual Warehouse:**
   - Also known as *data virtualization*.
   - A set of **views** over operational databases.

     As an *operational database* are all data sources considered that summarize, serve, and access up-to-date and real-time data. Generally, these are OLTP systems that provide ACID properties. These systems include, but are not limited to relational databases, NoSQL databases, but also unstructured data.
   - Only some of the possible summary views may be materialized.

# Extract, Transform, and Load (ETL)



- **Extract** Data:
  - Get data from multiple, heterogeneous, and external sources.
- **Clean** Data:
  - Detect errors in the data and rectify them if possible.
- **Transform** Data:
  - Convert data from legacy or host format to warehouse format.
- **Load** Data:
  - Sort, summarize, consolidate, compute views, check integrity, and build indexes and partitions.
- **Refresh** Data:
  - Propagate only the updates from the data sources to the warehouse.

# Metadata Repository[3]

Generally speaking:

**Metadata**

Data about data.

Three types: *business*, *process execution*, and *technical* metadata. **Business Metadata**

- Business terms and definitions.
- Logical data mapping.
- Data ownership.
- Charging policies.

**Process Execution Metadata**

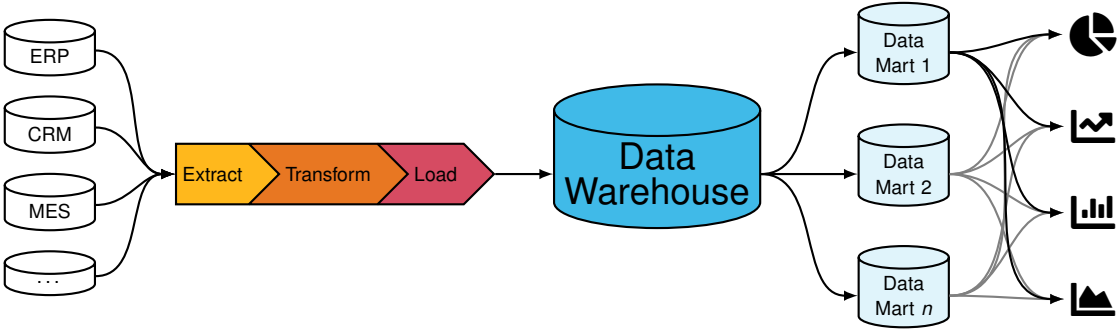
- Data acquisition schedule.
- Data-cleaning specifications.
- Aggregate specifications.
- Slowly changing dimensions policies.
- Duration of ETL / rows rejected and successful.

**Technical Metadata**

- Table structures and table attributes.
- Derived data definitions.
- Results from data profiling.
- Data lineage.

[3] C.f. chapter 9 of R. Kimball and J. Caserta, *The Data Warehouse ETL Toolkit: Practical Techniques for Extracting, Cleaning, Conforming, and Delivering Data*. Wiley, 2004, ISBN: 978-0764567575

# Data Warehouse Reference Overview



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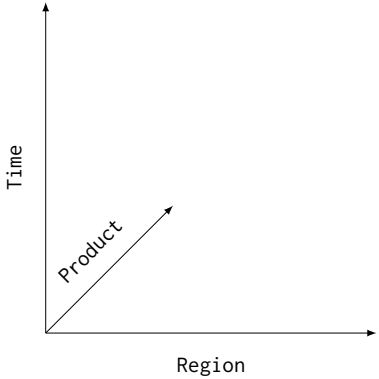
# Data Warehouse Modeling: Data Cube and OLAP

# Data Cubes



- Data warehouse is based on a **multidimensional data model**
- It views data in the form of a **data cube**.
- A **data cube** contains **two** different kinds of data:
  - **Dimensions:** Information that can be used to group the data.
    - A dimension often comes with different levels of granularity.
    - Example: Time (Granularity levels: day, month, quarter, year).
  - **Facts**: Information that can be aggregated.
    - Example: Price.

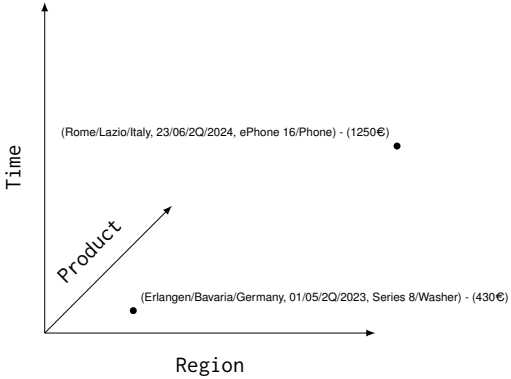
# Example: 3-D Data Cube




**Imagine:**

- 3-D coordinate system

# Example: 3-D Data Cube



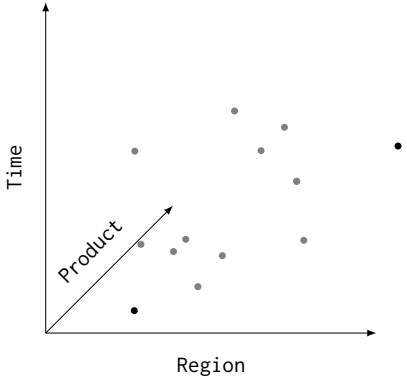

## Dimensional Values:

- Used to locate data points
- Example: Erlangen/Bavaria/Germany, 01/05/2Q/2023, Series 8/Washer

## Facts:

- Used to describe data points
- Example: 430€

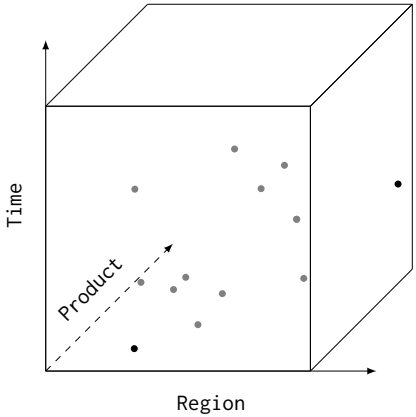
# Example: 3-D Data Cube




**A Real Data Cube:**

- Contains a lot of data points
- Many more than shown ...

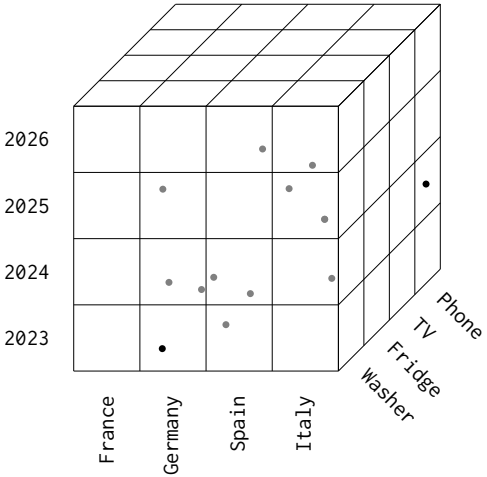
# Example: 3-D Data Cube




**The Data Cube:**

- Encapsulates all data points
- Can be used to aggregate all facts

# Example: 3-D Data Cube



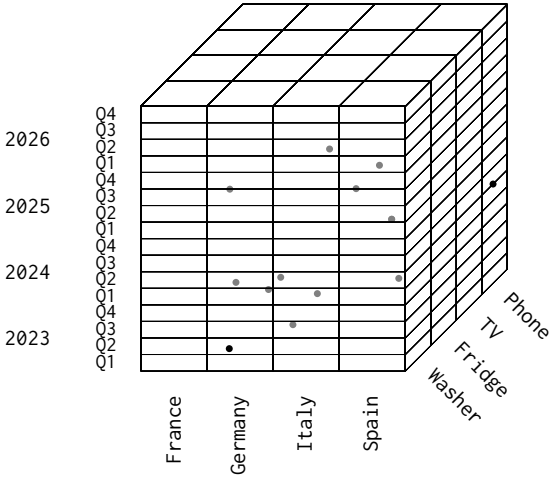

**A Cube of Cubes:**

- The cube can be sliced into smaller cubes

**The Smaller Cubes:**

- Can be analyzed separately
- Are in the highest granularity of each dimension

# Example: 3-D Data Cube

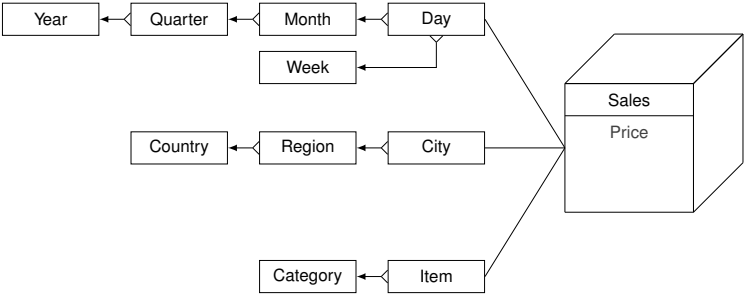



## Even Smaller Cubes:

- Often the cubes can be sliced into even smaller cubes
- By going to the next finer granularity level in at least one dimension

# mE/R Model of a Data Cube

- A **mE/R model**[5] contains both **dimensions** and **facts**.
- Very good in representing **dimensional hierarchies**.



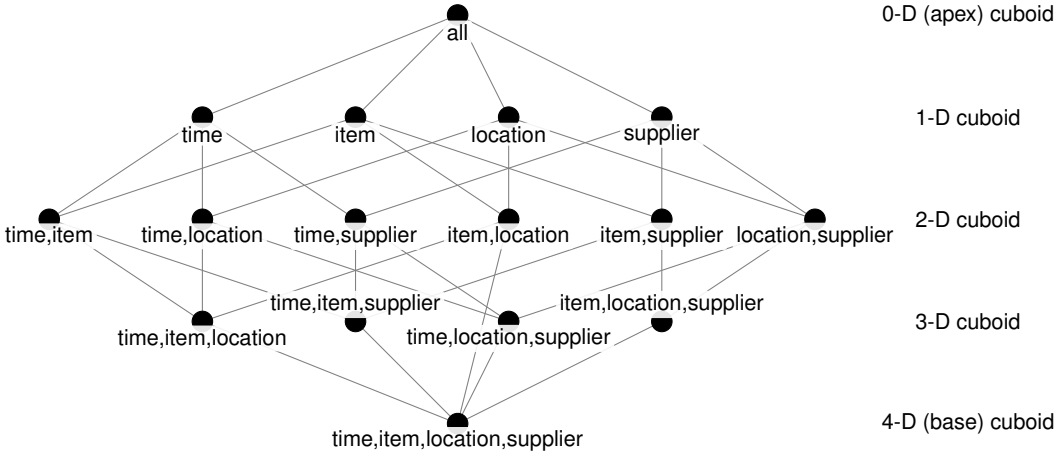

[5] C. Sapia et al., "Extending the E/R model for the multidimensional paradigm," in *Advances in Database Technologies, ER '98 Workshops on Data Warehousing and Data Mining, Mobile Data Access, and Collaborative Work Support and Spatio-Temporal Data Management, Singapore, November 19-20, 1998, Proceedings*, Y. Kambayashi et al., Eds., ser. Lecture Notes in Computer Science, vol. 1552, Springer, 1998, pp. 105–116. DOI: 10.1007/978-3-540-49121-7_9. [Online]. Available: https://doi.org/10.1007/978-3-540-49121-7%5C_9

# Narrowing the Data Cube Down



- Each data cube can be aggregated.
- In this process, it is possible to use only individual dimensions for aggregation:
  - ***n*-dimensional base cube.**
    - Called a base cuboid in data warehousing literature.
  - **Top most 0-dimensional cuboid.**
    - Holds the highest-level of summarization.
    - Called the apex cuboid.
  - **Lattice of cuboids.** (Forms a data cube)

# Cube: A Lattice of Cuboids

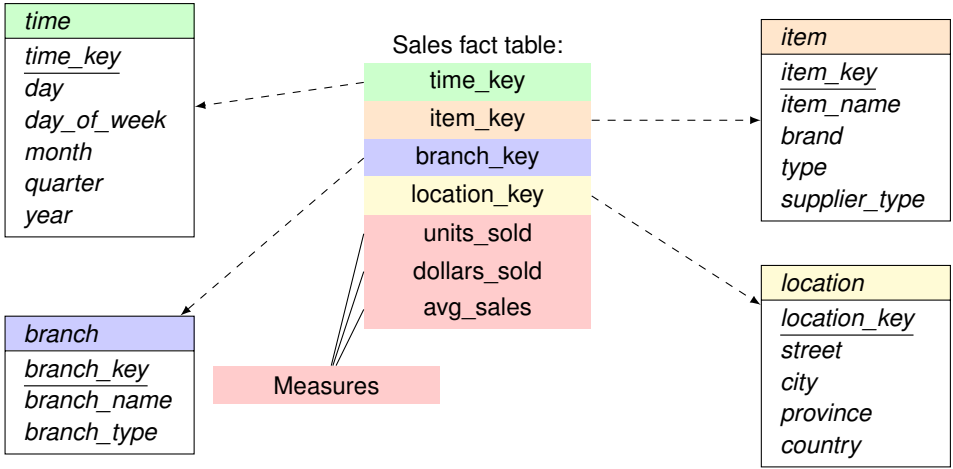


# Conceptual Modeling of Data Warehouses



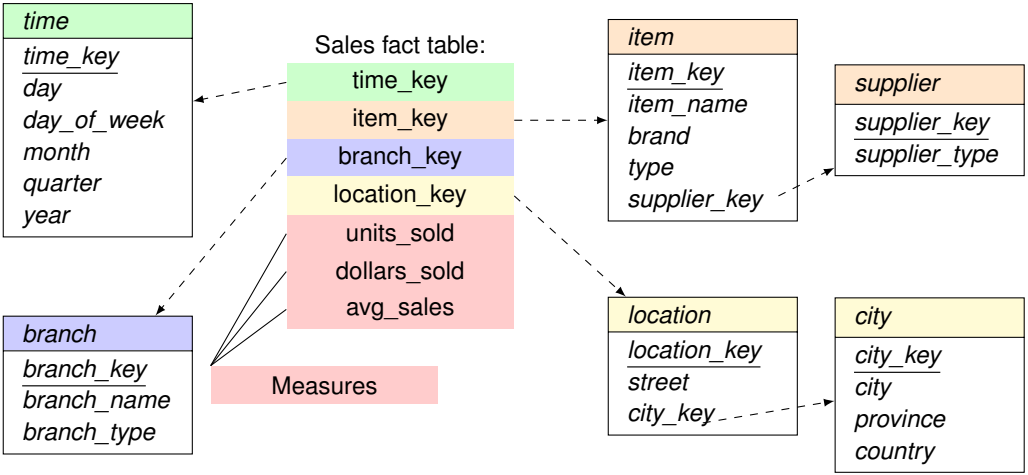
1. **Star schema:**.
   - A fact table in the middle connected to a set of dimension tables.
2. **Snowflake schema:**.
   - A refinement of the star schema where some dimensional hierarchies are **normalized** into a set of smaller dimension tables, forming a shape similar to a snowflake.
   - I. e. dimension tables of a star schema are split into multiple (dimension) tables along their respective granularity level, but not split/normalized for every granularity.
3. **Fact constellations:**.
   - Multiple fact tables sharing dimension tables, viewed as a collection of stars, therefore called **galaxy schema** or fact constellation.

# Example of a Star Schema

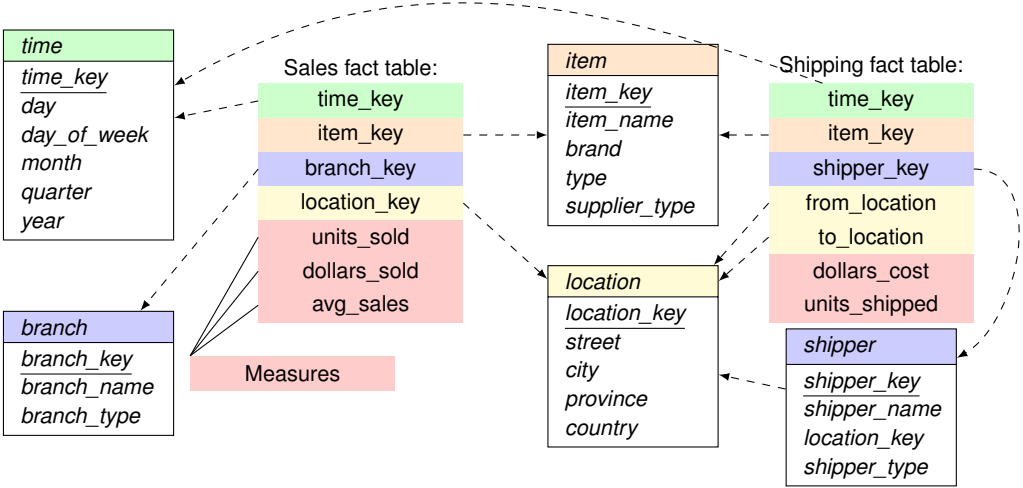


# Example of Snowflake Schema





# Example of Fact Constellation



# Data-Cube Measures



## Data-Cube Measure

A *data-cube measure* is a numeric function that can be evaluated at each point in the data cube space.

**Three Categories:**

1. **Distributive:**
   - If the result derived by applying the function to the $n$ aggregate values obtained for $n$ partitions of the dataset is the same as that derived by applying the function on all the data without partitioning.
     E.g. `COUNT, SUM, MIN, MAX`.
2. **Algebraic:**
   - If it can be computed by an algebraic function with $M$ arguments, each of which is obtained by applying a distributive aggregate function.
     E.g. `AVG`, $\texttt{MIN}_N$, `STD`.
3. **Holistic:**
   - If there is no constant bound on the storage size needed to describe a subaggregate.
     E.g. `MEDIAN, MODE, RANK`.

# Aggregation Type



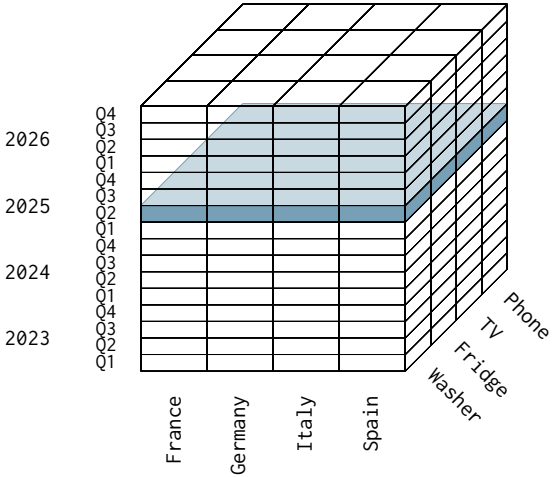
- **Non-trivial property.**
  - Next to name and value range.
- **Defines the set of aggregation operations that can be executed on a measure (a fact).**
- **STOCK:** Measure at a specific point in time.
  - Aggregated as desired.
  - E.g. sales turnover, quantity of an item ordered per day.
- **FLOW:** Measure over a period of time.
  - Aggregated as desired, but temporal aggregation not permitted.
  - E.g. total stock and total inventory. Yet, summarization of article stock over multiple days makes no sense!
- **VPU (Value per Unit):** Measures that cannot be summed.
  - E.g. unit price, tax rates, exchange rates.
- **Always applicable: `MIN`, `MAX` and `AVG`.**

# Typical OLAP Operations



- **Slice and dice: project and select.**
  - Selecting only certain dimensions/value ranges from a cube
- **Roll up (drill up): summarize data.**
  - By climbing up hierarchy or by dimension reduction.
- **Drill down (roll down): reverse of roll up.**
  - From higher-level summary to lower-level summary or detailed data, or introducing new dimensions.
- **Pivot (rotate):**
  - Reorient the cube, visualization, 3D to series of 2D planes.
- **Other operations:**
  - **Drill across:** involving (across) more than one fact table.
  - **Drill through:** through the bottom level of the cube to its back-end relational tables (using SQL).

# OLAP Operation: Slice




## Slice

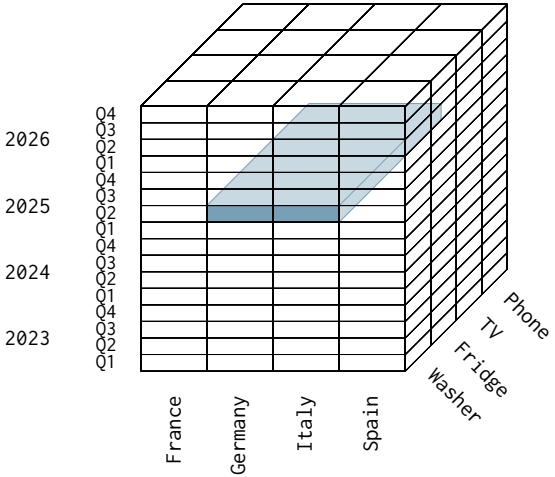
Basic idea:

- Perform a selection on one dimension of the cube.

Example:

- Select the data for Q2 2025.

# OLAP Operation: Dice




## Dice

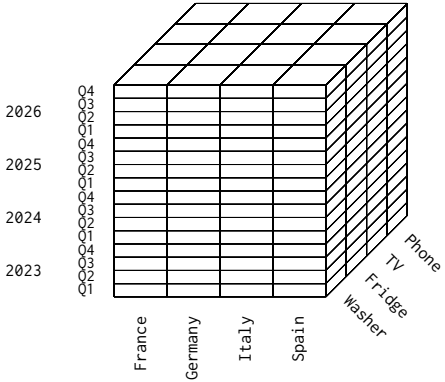
Basic idea:

- Perform a selection on more than one dimension.
- It does not have to be on all dimensions!

Example:

- Select the data for Q2 2025 and the regions Germany and Italy.

# OLAP Operations: Roll Up and Drill Down

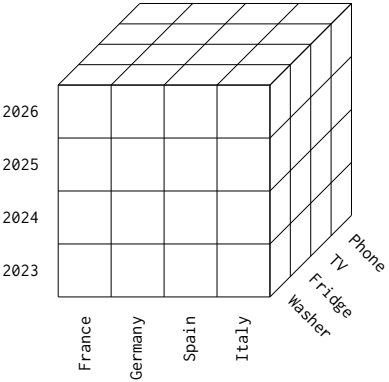



**Roll Up**

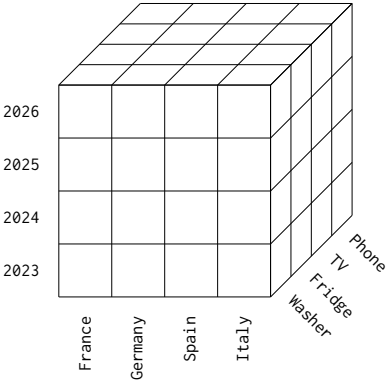
Switch to a coarser granularity.

**Drill Down**

Switch to a finer granularity.

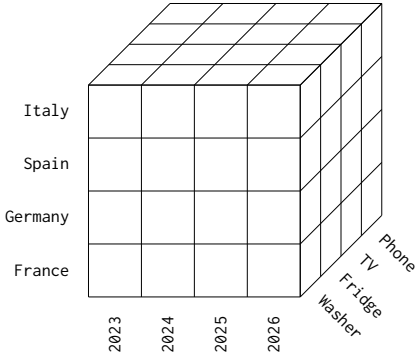


# OLAP Operations: Pivot




**Pivot**

Rotate the cube to provide a different view on the data.



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# Data Warehouse Design and Usage

# Design of Data Warehouse: A Framework



**Four views regarding the design of a data warehouse:**

- **Top-down view:**
  - Allows selection of the relevant information necessary for the data warehouse.
- **Data-source view:**
  - Exposes the information being captured, stored, and managed by operational systems.
- **Data warehouse view:**
  - Consists of fact tables and dimension tables.
- **Business-query view:**
  - Sees the perspectives of data in the warehouse from the view of the end-user.

# Data Warehouse Design Process

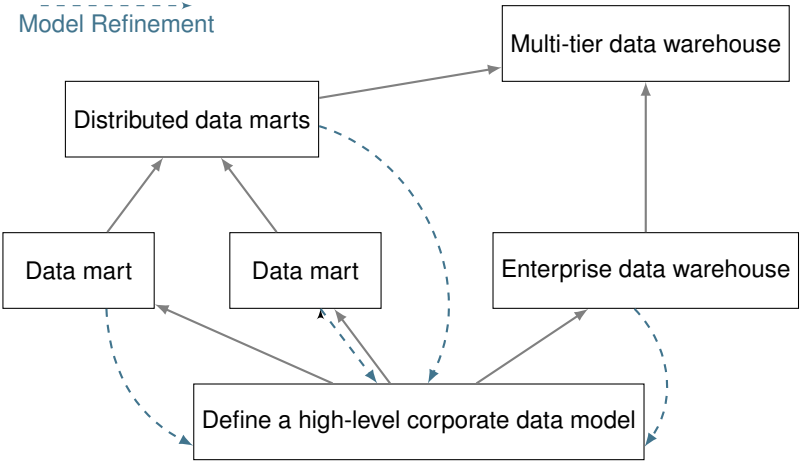


- **Top-down, bottom-up approaches or a combination of both:**
  - **Top-down:** starts with overall design and planning (mature).
  - **Bottom-up:** starts with experiments and prototypes (rapid).
- **From software-engineering point of view:**
  - **Waterfall:** structured and systematic analysis at each step before proceeding to the next.
  - **Spiral:** rapid generation of increasingly functional systems, short turn-around time.
- **Typical Data warehouse design process:**
  1. Choose a **business process** to model, e.g., orders, invoices, etc.
  2. Choose a **grain** (atomic level of data) of the business process.
  3. Choose **dimensions** that will apply to each fact-table record.
  4. Choose a **measure** that will populate each fact-table record.

## DWH Construction is No Easy Feat

Construction of a data warehouse is a difficult long-term task. It is absolutely necessary that its implementation scope is clearly defined at the beginning. Goals and tasks should be *SMART* (specific, measurable, achievable, relevant, and time-related).

# Data Warehouse Development



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# Data Warehouse and Data Mining

# From OLAP To OLAM[5]

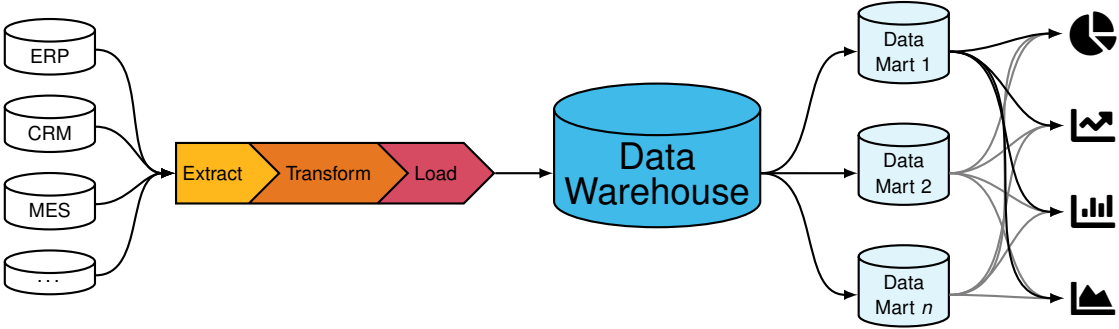


**Why online analytical mining?**

- DW contains integrated, consistent, cleaned data.
- Available information-processing structure surrounding data warehouses.
  - ODBC, OLEDB, Web access, service facilities, reporting, and OLAP tools.
- OLAP-based exploratory data analysis.
  - Mining with drilling, dicing, pivoting, etc.
- Online selection of data-mining functions.
  - Integration and swapping of multiple mining functions, algorithms, and tasks.

[5] OLAM = Online Analytical Mining

# Data Warehouse and Data Mining




- Data mining algorithms in transformation step: E.g. integrate articles from two systems that have different article group hierarchy. Goal: Map one article group hierarchy to the existing article group hierarchy.
- Frequent pattern mining and clustering in reporting: E.g. affinity analysis, revenue prediction, cluster customers and use this insight for a new marketing campaign.

# Summary

# Summary



- **Data warehousing: multi-dimensional model of data.**
  - A data cube consists of dimensions and measures.
  - Star schema, snowflake schema, fact constellations.
  - OLAP operations: drilling, rolling, slicing, dicing and pivoting.
- **Data warehouse architecture, design, and usage.**
  - Multi-tiered architecture.
  - Business-analysis design framework.
  - Information processing, analytical processing, data mining, OLAM (Online Analytical Mining).

# Interested in More Information?

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Our appendix in this document covers:

- **Implementation: efficient computation of data cubes.**
  - Partial vs. full vs. no materialization.
  - Indexing OLAP data: Bitmap index and join index.
  - OLAP query processing.
  - OLAP servers: ROLAP, MOLAP, HOLAP.
- **Data generalization: attribute-oriented induction.**

Additionally, check out these books:


- R. Kimball and J. Caserta, *The Data Warehouse ETL Toolkit: Practical Techniques for Extracting, Cleaning, Conforming, and Delivering Data*. Wiley, 2004, ISBN: 978-0764567575
- W. H. Inmon, *Building the Data Warehouse*. Wiley, 2005, 4th edition, ISBN: 978-076459-944-6
- In German: A. Bauer and H. Günzel, *Data Warehouse Systeme – Architektur, Entwicklung, Anwendung*. dpunkt.verlag GmbH, 2004, 4th edition, ISBN: 978-3-89864-785-4

**Any questions about this chapter?**

Ask them now or ask them later in our forum:

https://www.studon.fau.de/studon/goto.php?target=lcode_OLYeD79h

# Appendix

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# Data Warehouse Implementation

# Efficient Data-Cube Computation



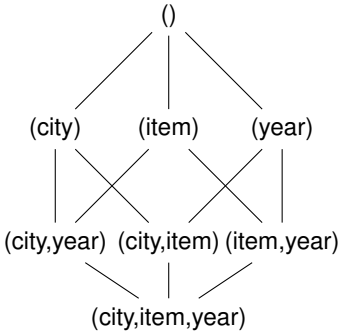
- **Data cube can be viewed as a lattice of cuboids.**
- The bottom-most cuboid is the base cuboid.
- The top-most cuboid (apex) contains only one cell.
- How many cuboids in an $n$-dimensional cube with $L_i$ levels associated with dimension $i$?

$$T = \prod_{i=1}^{n} (L_i + 1). \tag{1}$$

- **Materialization of data cube.**
  - Materialize each (cuboid) (full materialization), none (no materialization), or some (partial materialization).
  - Selection of cuboids to materialize based on size, sharing, access frequency, etc.

# The "Compute Cube" Operator

- **Cube definition and computation in DMQL:**
  `DEFINE CUBE sales [item, city, year]:`
  `SUM (sales_in_dollars);`
  `COMPUTE CUBE sales;`
- **Transform it into an SQL-like language:**
  with a new operator `CUBE BY` (Gray et al. 96).
  `SELECT item, city, year, SUM (amount)`
  `FROM sales`
  `CUBE BY item, city, year;`
- **Need to compute the following** `GROUP BY`**s:**
  `(city, item, year),`
  `(city, item), (city, year),`
  `(item, year),`
  `(city), (item), (year)`
  `()`



# Indexing OLAP Data: Bitmap Index



- **Index on a particular column.**
- **Each value in the column has a bit vector: bit-op is fast.**
- **Length of bit vector: # of records in base table.**
- ***i*-th bit set, if *i*-th row of base table has value of bit vector.**
- **Not suitable for high-cardinality domains:**
  - A bit compression technique called Word-Aligned Hybrid (WAH) makes it work for high-cardinality domain as well [Wu et al., TODS'06].

**Base table**

| Cust | Region | Type |
|---|---|---|
| $C1$ | Asia | Retail |
| $C2$ | Europe | Dealer |
| $C3$ | Asia | Dealer |
| $C4$ | America | Retail |
| $C5$ | Europe | Dealer |

**Index on region**

| RecID | Asia | Europe | America |
|---|---|---|---|
| 1 | 1 | 0 | 0 |
| 2 | 0 | 1 | 0 |
| 3 | 1 | 0 | 0 |
| 4 | 0 | 0 | 1 |
| 5 | 0 | 1 | 0 |

**Index on type**

| RecID | Retail | Dealer |
|---|---|---|
| 1 | 1 | 0 |
| 2 | 0 | 1 |
| 3 | 0 | 1 |
| 4 | 1 | 0 |
| 5 | 0 | 1 |

# Indexing OLAP Data: Join Indices

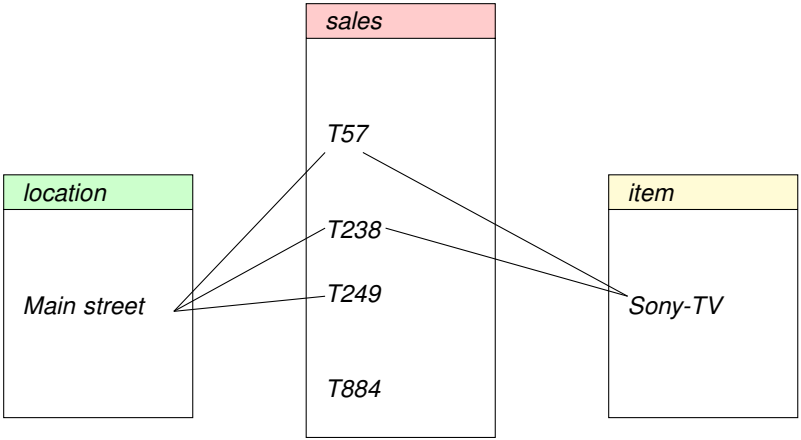


- **Join index:**

$$\mathrm{JI}(\texttt{R-id}, \texttt{S-id}) \quad \text{where} \quad \mathrm{R}(\texttt{R-id}, \ldots) \bowtie \mathrm{S}(\texttt{S-id}, \ldots). \tag{2}$$

- **Traditional indices map the values to a list of record ids.**
  - Materializes relational join in JI-file and speeds it up.
- **In data warehouses, join index relates the values of the dimensions of a star schema to rows in the fact table.**
  - E.g. fact table: Sales and two dimensions location and item.
    - A join index on location maintains for each distinct location a list of `R-ids` of the tuples recording the sales in that location.
  - Join indices can span multiple dimensions.

# Indexing OLAP data: Join Indices (Example)



# Efficient Processing of OLAP Queries



- **Determine which operations should be performed on the available cuboids.**
  - Transform drill, roll, etc. into corresponding SQL and/or OLAP operations.
    E.g. `dice` = `selection` + `projection`.
- **Determine which materialized cuboid(s) should be selected for OLAP operation.**
  - Let the query to be processed be on `{brand, province_or_state}` with the condition "`year = 2004`", and there are 4 materialized cuboids available:
    1) `year, item_name, city`.
    2) `year, brand, country`.
    3) `year, brand, province_or_state`.
    4) `item_name, province_or_state where year = 2004`.
  - Which should be selected to process the query?
- **Explore indexing structures and compressed vs. dense-array structures in MOLAP.**

# OLAP Server Architectures



- **Relational OLAP (ROLAP).**
  - Use relational or extended-relational DBMS to store and manage warehouse data and OLAP middleware.
  - Include optimization of DBMS backend, implementation of aggregation navigation logic, and additional tools and services.
  - Greater scalability.
- **Multidimensional OLAP (MOLAP).**
  - Sparse array-based multidimensional storage engine.
  - Fast indexing to pre-computed summarized data.
- **Hybrid OLAP (HOLAP) (e.g., Microsoft SQL-Server).**
  - Flexibility, e.g., low level: relational, high-level: array.
- **Specialized SQL servers (e.g., Redbricks).**
  - Specialized support for SQL queries over star/snowflake schemas.

# Attribute-Oriented Induction

# Data Generalization



- **Summarize data:**
  - **By replacing relatively low-level values**
    e.g. numerical values for the attribute `age`
    **with higher-level concepts**
    e.g. `young`, `middle-aged` and `senior`.
  - **By reducing the number of dimensions**
    e.g. removing `birth_date` and `telephone_number`
    when summarizing the behavior of a group of students.
  - Describe concepts in concise and succinct terms at generalized (rather than low) levels of abstractions:
    - Facilitates users in examining the general behavior of the data.
    - Makes dimensions of a data cube easier to grasp.

# Attribute-Oriented Induction



- **Proposed in 1989** (KDD'89 workshop).
- **Not confined to categorical data nor to particular measures.**
- **How is it done?**
  - Collect the **task-relevant data** (initial relation) using a relational database query.
  - Perform **generalization** by attribute removal or attribute generalization.
  - Apply **aggregation** by merging identical, generalized tuples and accumulating their respective counts.
  - Interaction with users for knowledge presentation.

# Attribute-Oriented Induction: An Example



- **Example:** Describe general characteristics of graduate students in a university database.
- **Step 1:** Fetch relevant set of data using an SQL statement, e.g.

```
SELECT name, gender, major, birth_place, birth_date, residence, phone#, gpa
FROM student
WHERE student_status IN "Msc", "MBA", "PhD";
```

- **Step 2:** Perform attribute-oriented induction.
- **Step 3:** Present results in generalized-relation, cross-tab, or rule forms.

# Class Characterization (I)

| Name | Gender | Major | Birth place | Birth date | Residence | Phone number | GPA |
|---|---|---|---|---|---|---|---|
| Jim | M | CS | Vancouver, BC, Canada | 08-21-76 | 3511 Main St., Richmond | 687-4598 | 3.67 |
| Scott Lachance | M | CS | Montreal, Que, Canada | 28-07-75 | 345 1st Ave., Richmond | 253-9106 | 3.70 |
| Laura Lee | F | Physics | Seattle, WA, USA | 25-08-70 | 125 Austin Ave., Burnaby | 420-5232 | 3.83 |
| Removed | Retained | Sci, Eng, Bus | Canada, Foreign | Age range | City | Removed | Excl, Vg,. . . |

# Class Characterization (II)

| Gender | Major | Birth region | Age range | Residence | GPA | Count |
|---|---|---|---|---|---|---|
| M | Science | Canada | 20-25 | Richmond | Very good | 16 |
| F | Science | Foreign | 25-30 | Burnaby | Excellent | 22 |
| ... | ... | ... | ... | ... | ... | ... |

# Class Characterization (III)

Cross-table of birth region and gender:

| | Canada | Foreign | Total |
|---|---|---|---|
| M | 16 | 14 | 30 |
| F | 10 | 22 | 32 |
| Total | 26 | 36 | 62 |

# Basic Principles of Attribute-Oriented Induction



- **Data focusing:**
  - Task-relevant data, including dimensions.
  - The result is the **initial relation**.
- **Attribute removal:**
  - Remove attribute A, if there is a large set of distinct values for A,
    but (1) there is no generalization operator on A,
    or (2) A's higher-level concepts are expressed in terms of other attributes.
- **Attribute generalization:**
  - If there is a large set of distinct values for A,
    and there exists a **set of generalization operators** on A,
    then select an operator and generalize A.
- **Attribute-threshold control:**
  - Typical 2-8, specified/default.
- **Generalized-relation-threshold control:**
  - Control the final relation/rule size.

# Attribute-Oriented Induction: Basic Algorithm



- **InitialRel:**
  - Query processing of task-relevant data, deriving the initial relation.
- **PreGen:**
  - Based on the analysis of the number of distinct values in each attribute, determine generalization plan for each attribute: removal? Or how high to generalize?
- **PrimeGen:**
  - Based on the PreGen plan, perform generalization to the right level to derive a "prime generalized relation", accumulating the counts.
- **Presentation:**
  - User interaction:
    1. Adjust levels by drilling.
    2. Pivoting.
    3. Mapping into rules, cross tabs, visualization presentations.

# Presentation of Generalized Results



- **Generalized relation:**
  - Relations where some or all attributes are generalized, with counts or other aggregation values accumulated.
- **Cross tabulation:**
  - Mapping results into cross-tabulation form (similar to contingency tables).
  - Visualization techniques: pie charts, bar charts, curves, cubes, and other visual forms.
- **Quantitative characteristic rules:**
  - Mapping generalized results into characteristic rules with quantitative information associated with it, e.g.

$$\text{grad}(x) \wedge \text{male}(x) \implies \tag{3}$$

$$\text{birth_region}(x) = \text{"Canada"}[t : 53\%] \vee \tag{4}$$

$$\text{birth_region}(x) = \text{"foreign"}[t : 47\%]. \tag{5}$$

# Mining-Class Comparisons



- **Comparison: Comparing two or more classes.**
- **Method:**
  - Partition the set of relevant data into the **target class** and the **contrasting class(es)**.
  - Generalize both classes to the same high-level concepts (i.e. AOI).
    - Including aggregation.
  - Compare tuples with the same high-level concepts.
  - Present for each tuple its description and two measures.
    - Support – distribution within single class (counts, percentage).
    - Comparison – distribution between classes.
  - Highlight the tuples with strong discriminant features.
- **Relevance Analysis:**
  - Find attributes (features) which best distinguish different classes.

# Attribute-Oriented Induction vs. OLAP



- **Similarity:**
  - Data generalization.
  - Presentation of data summarization at multiple levels of abstraction.
  - Interactive drilling, pivoting, slicing and dicing.
- **Differences:**
  - OLAP has systematic preprocessing, query independent, and can drill down to rather low level.
  - AOI has automated desired-level allocation and may perform dimension-relevance analysis/ranking when there are many relevant dimensions.
  - AOI works on data which are not in relational forms.